

Where It Breaks

Attempts to Break the Website

Carried out by myself and Latifah Bashir (peer-review)

#	Break attempt	Result
1	Submit the form empty	The browser puts an indicator next to the required field saying "Please fill out this field". The first required box glows gold. Nothing is submitted.
2	Enter any string in the problem/input field	Any string is allowed. The field just looks for text and does not restrict what is entered.
3	Submit twice (click button multiple times)	The form does not allow a second submission. The page has to be reloaded before another submission is possible.
4	Submit with a very long problem input (10k–15k characters)	The form still accepts the submission with no visible character limit.
5	Reload after entering content	Reloading keeps the content in the fields, but the form can then be submitted again.
6	Use spaces in every field	The name field accepts spaces. The email field correctly asks for an email address. The problem and name fields allow the input.
7	Use emoji-only input	Emoji-only input is accepted in the name and problem fields. The email field correctly asks for an email address.
8	Enter <code><script>alert(1)</script></code>	The email field refuses the input. The other fields accept it as text. No script executes.
9	Enter a@b as an email	a@b is accepted as a valid email.
10	Enter not-an-email as an email	The email field correctly refuses it.
11	Test in different browsers	Opened the site in Microsoft Edge and Chrome after previously using Samsung Internet and Firefox. Everything worked: links, images, resume, sizing, and overall look.
12	Check images/GIFs on phone	GIF and image quality looked fine on a phone screen.
13	Click everything	Tested every nav item, footer link, case study, "Read the case study", both "Email me directly" mailto links, the CV download, and all repo/demo links in incognito. Everything worked.
14	Double-click Send / press Enter twice	Only the first action does anything. Subsequent clicks/Enter presses do

		nothing. No duplicate submission was created.
15	Submit with no internet connection	The site shows “message couldn’t send / something went wrong” with an option to try again or email directly. When the connection came back, the message sent successfully.
16	Try to create duplicate messages through repeated online/offline attempts	No duplicate messages were sent. I tried switching the internet on and off and clicking multiple times through several iterations. Only the first successful click sent the email.

Triage

Fix Now

Finding	Why it is a fix-now
#2 Any string is allowed in the problem field	This is not necessarily a failure by itself, but the input has no meaningful validation beyond requiring text. This could allow very low-quality or nonsensical submissions.
#4 No character limit on the problem field	The form accepted 10k–15k characters. A practical limit would prevent unnecessarily large submissions and protect the form from excessive input.
#9 a@b is accepted as an email	The email passes the browser's basic email validation even though it is not a realistic email address. Stronger validation could improve input quality.

Known Limitations

Finding	Why it is a known limitation
#1 Empty form submission	The browser's built-in required-field validation handles this correctly. Nothing is submitted.
#3 Cannot submit twice without reload	This is intentional protection against duplicate submissions.
#6 Spaces accepted in name/problem fields	These fields are text inputs, so accepting spaces is expected.
#5 Reload after entering content keeps content	Keeping this functionality to allow for reloading/losing internet connection and content being retained. Moving to different page still refreshes the content.
#7 Emoji-only input accepted	The fields accept text, and emoji are valid text characters. This is not a functional break.

#8 Script string accepted in non-email fields	The string is treated as text and does not execute as a script, so there was no actual XSS break observed.
#10 Invalid email refused	This is expected validation behaviour, not a problem.
#11 Browser compatibility	Tested successfully across Samsung Internet, Firefox, Microsoft Edge, and Chrome. No browser-specific break was found.
#12 Mobile image/GIF quality	Images and GIFs remained clear and usable on a phone screen.
#13 Links and interactive elements	Everything tested worked, including navigation, case studies, email links, CV download, and repo/demo links.
#14 Double-click / Enter twice	The site prevents repeated submission, which is desirable behaviour.
#15 Offline submission	The site cannot send a message without an internet connection. The user receives an error and an alternative way to contact me.
#16 Duplicate emails	Multiple attempts did not create duplicate messages. The submission protection worked as intended.

Note: the known limitation regarding the text input are accepted due to the minimal risk. The cost is an erroneous submission being forwarded to my mail. No post processing is being done on these results so as it stands most limitations are acceptable for the use-case. Extensive validation is not necessary at this juncture.

Overall Audit Result

The main weaknesses I found were around **input validation and input length**, rather than broken navigation, browser compatibility, mobile display, or duplicate submissions. The more serious interaction risks I tested (rapid repeated submission, repeated Enter presses, and online/offline switching) did **not** produce duplicate messages/break the site.

Evidence That Fix-Nows Were Addressed

All fixes are confirmed live on deployed website

#2 Any string allowed in the problem field

Before:

The problem field accepted any string as long as text was entered. Inputs such as asdf or a wall of repeated characters could be submitted.

Fix applied:

- The problem field now requires **at least 20 characters and two or more words**. Inputs that do not meet that requirement return: *"That's too short to work from. Describe the problem in a sentence or two."*
- The name field was also tightened so it requires **at least 2 characters**, preventing a lone character from being submitted.

Evidence:

Retested the previously weak inputs after the change. Single-word/low-value inputs are now rejected while normal sentence-length problem descriptions are accepted.

Status: Fixed.

#4 No character limit on the problem field

Before:

The problem field accepted extremely large inputs, including **10,000–15,000 characters**.

Fix applied:

- The textarea now has a hard **maxlength="2000"** enforced by the browser, including with JavaScript disabled.
- A live **x / 2,000 characters** counter was also added below the field. The counter changes to gold as the user approaches the limit, making the constraint visible before they reach it.
- The name and company fields also received **100-character limits** for consistency.

Evidence:

Retested with long input. The field stops accepting input at 2,000 characters rather than allowing an unrestricted 10k–15k character submission. The visible counter also shows the user how much space remains.

Status: Fixed.

#9 a@b accepted as a valid email

Before:

The browser accepted a@b as a valid email address.

Fix applied:

Email validation was strengthened so the address must contain a **dot followed by a 2+ letter TLD**.

Examples now behave as follows:

- a@b → rejected
- a@b.c → rejected
- a@b.co → accepted
- her@company.com → accepted

The existing friendly validation message is retained.

Evidence:

Retested the original failing input a@b and the additional edge case a@b.c. Both are now rejected, while realistic email formats continue to pass.

Status: Fixed.

Screenshot evidence of applied fixes

#2 Any string allowed in the problem field

Before vs After

YOUR NAME
Gabby

YOUR EMAIL
gabby31055@deposin.com

COMPANY (OPTIONAL)

THE PROBLEM YOU'D LIKE SOLVED
Short problem

Sending...

Submissions land in my inbox. I reply from theresa.saumu@gmail.com.

Got it.

Your message is on its way to my inbox. I'll reply from theresa.saumu@gmail.com.

vs

YOUR EMAIL
gabbymartin@gmail.com

COMPANY (OPTIONAL)

THE PROBLEM YOU'D LIKE SOLVED
Short Problem

15 / 1,000 characters

Send the problem →

Submissions land in my inbox. I reply from theresa.saumu@gmail.com.

That's too short to work from. Describe the problem in a sentence or two.

Email me directly → LinkedIn GitHub →

#4 No character limit on the problem field

YOUR NAME
Gabby

YOUR EMAIL
gabby31055@deposin.com

COMPANY (OPTIONAL)

THE PROBLEM YOU'D LIKE SOLVED
nisi quis portitor congue. elit erat euismod orci. ac placerat dolor lectus quis orci. Phasellus consectetur vestibulum elit. Aenean tellus metus, bibendum sed, posuere ac, mattis non, nunc. Vestibulum fringilla pede sit amet augue. In turpis. Pellentesque posuere. Prossent turpis. Aenean posuere, tortor sed cursus feugiat, nunc augue blandi nunc. eu sollicitudin urna dolor sagittis lacus. Donec elit libero, sodales nec, volutpat a, suscipit non, turpis.

Sending...

Submissions land in my inbox. I reply from theresa.saumu@gmail.com.

Got it.

Your message is on its way to my inbox. I'll reply from theresa.saumu@gmail.com.

Email me directly → LinkedIn GitHub →

vs

YOUR EMAIL
gabbymartin@gmail.com

COMPANY (OPTIONAL)

THE PROBLEM YOU'D LIKE SOLVED
ut metus varius laoreet. Quisque rutrum. Aenean imperdiet. Etiam ultricies nisi vel augue. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maec

2,300 / 1,000 characters

Sent

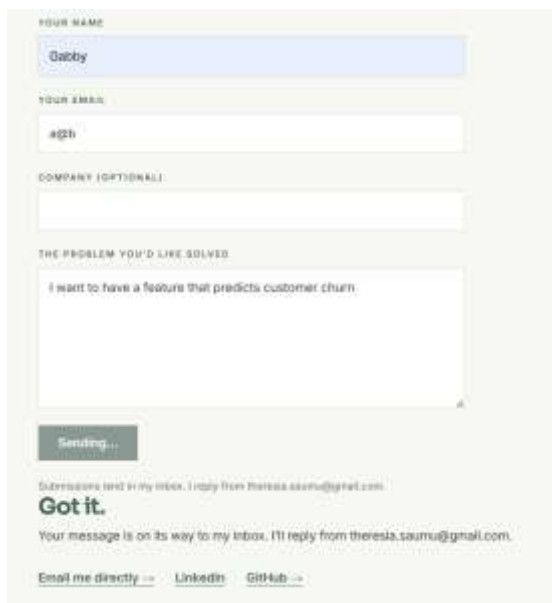
Submissions land in my inbox. I reply from theresa.saumu@gmail.com.

Thanks, message sent.

I'll be in touch from theresa.saumu@gmail.com.

Email me directly → LinkedIn GitHub →

#9 a@b accepted as a valid email



YOUR NAME
Gabby

YOUR EMAIL
a@b

COMPANY (OPTIONAL)

THE PROBLEM YOU'D LIKE SOLVED
I want to have a feature that predicts customer churn.

Sending...

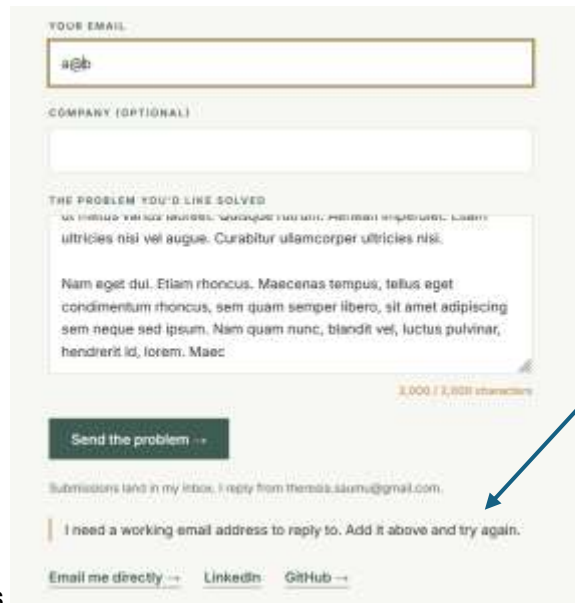
Submissions sent to my inbox. I reply from theresa.saumu@gmail.com.

Got it.

Your message is on its way to my inbox. I'll reply from theresa.saumu@gmail.com.

Email me directly → LinkedIn GitHub →

VS



YOUR EMAIL
a@b

COMPANY (OPTIONAL)

THE PROBLEM YOU'D LIKE SOLVED
ut tristique vel non. Curabitur ullamcorper ultricies nisi. Nam eget dui. Etiam rhoncus. Maecenas tempus, tellus eget condimentum rhoncus, sem quam semper libero, sit amet adipiscing sem neque sed ipsum. Nam quam nunc, blandit vel, luctus pulvinar, hendrerit id, lorem. Maec

2,000 / 2,000 characters

Send the problem →

Submissions sent to my inbox. I reply from theresa.saumu@gmail.com.

I need a working email address to reply to. Add it above and try again.

Email me directly → LinkedIn GitHub →

SEO Added

1. **Page titles:**
 - a. Home: Theresia Saumu | Applied AI & ML Engineer
 - b. Contact: Contact | Theresia Saumu
 - c. Agentic BI Builder: Agentic BI Builder | Theresia Saumu
 - d. Financial Risk Monitor: Financial Risk Monitor | Theresia Saumu
 - e. SolarSense: SolarSense | Theresia Saumu
 - f. StudyGapAI: StudyGapAI | Theresia Saumu
 - g. Alzheimer's Disease Classification: Alzheimer's Classification | Theresia Saumu
2. **Description:** homepage: "Portfolio of Theresia Saumu, an Applied AI and Machine Learning engineer building useful systems with AI, ML, data, analytics, and automation"; each page has a matching unique one.
3. **Social-share preview:** Open Graph + Twitter card tags pointing at a new 1200×630 social-card.jpg (54 KB): your portrait blended into the brand green, name in cream, "Applied AI & ML Engineer" in gold. This is what renders when you paste the URL into LinkedIn, WhatsApp, Slack, or X. After merge, test it at.opengraph.xyz or LinkedIn's Post Inspector (LinkedIn caches aggressively, the inspector clears it).
4. Plumbing - **canonical URLs** on every page, an *auto-generated sitemap-index.xml*, a *robots.txt* pointing crawlers at it, and the branded 404.